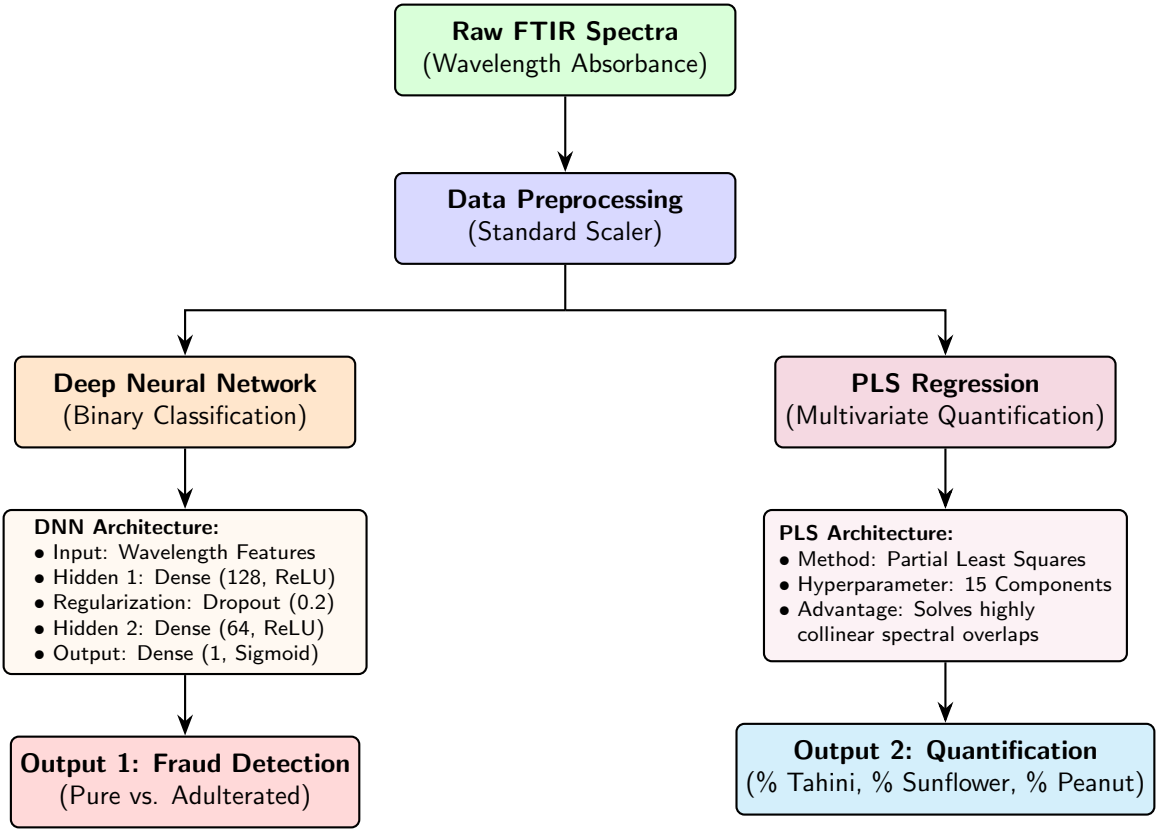
Raw FTIR Spectra
(Wavelength Absorbance)
Data Preprocessing
(Standard Scaler)
Deep Neural Network
(Binary Classification)
PLS Regression
(Multivariate Quantification)
DNN Architecture:
• Input: Wavelength Features
• Hidden 1: Dense (128, ReLU)
• Regularization: Dropout (0.2)
• Hidden 2: Dense (64, ReLU)
• Output: Dense (1, Sigmoid)
PLS Architecture:
• Method: Partial Least Squares
• Hyperparameter: 15 Components
• Advantage: Solves highly collinear spectral overlaps
Output 1: Fraud Detection
(Pure vs. Adulterated)
Output 2: Quantification
(% Tahini, % Sunflower, % Peanut)